



Teacher: **Yanjie Li**
Course: **Linear Algebra in Control Theory**

Assignment Number: **3**
Disclosure date: June 29, 2026

Problem 1

In $\mathbf{R}^4$, let

$$U = \text{span}((1, 1, 0, 0), (1, 1, 1, 2)).$$

Find $u \in U$ such that $\|u - (1, 2, 3, 4)\|$ is as small as possible.

Problem 2

Find $p \in \mathcal{P}_3(\mathbf{R})$ such that $p(0) = 0, p'(0) = 0$, and

$$\int_0^1 |2 + 3x - p(x)|^2 dx$$

is as small as possible.

Problem 3

Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V .

- (a) Prove that if $U \subset \text{null } T$, then U is invariant under T .
- (b) Prove that if $\text{range } T \subset U$, then U is invariant under T .

Problem 4

Suppose $S, T \in \mathcal{L}(V)$ are such that $ST = TS$. Prove that $\text{range } S$ is invariant under T .

Problem 5

Suppose $S, T \in \mathcal{L}(V)$ are such that $ST = TS$. Prove that null S is invariant under T .

Problem 6

Define $T \in \mathcal{L}(\mathbf{F}^3)$ by

$$T(z_1, z_2, z_3) = (2z_2, 0, 5z_3)$$

Find all eigenvalues and eigenvectors of T .

Problem 7

Define $T : \mathcal{P}(\mathbf{R}) \rightarrow \mathcal{P}(\mathbf{R})$ by $Tp = p'$. Find all eigenvalues and eigenvectors of T .

Problem 8

Suppose $T \in \mathcal{L}(V)$. Suppose $S \in \mathcal{L}(V)$ is invertible. (a) Prove that T and $S^{-1}TS$ have the same eigenvalues. (b) What is the relationship between the eigenvectors of T and the eigenvectors of $S^{-1}TS$?

Problem 9

Find all eigenvalues and eigenvectors of the backward shift operator $T \in \mathcal{L}(\mathbf{F}^\infty)$ defined by

$$T(z_1, z_2, z_3, \dots) = (z_2, z_3, \dots).$$

Problem 10

If A is a matrix with $m \times n$ dimension, please show that $A^T A$ and AA^T have the same nonzero eigenvalues.

Pay Attention

- a) Mark your class number, student number and name on the homework.
- b) Try to write your homework on **A4** size paper.
- c) Please hand in your homework to your TA before class next Monday (July 6).

References

- [1] Axler, S. (1997). Linear algebra done right. Springer Science & Business Media.
- [2] Lay, D. C. . Linear algebra and its applications. Academic Press.
- [3] Leon, S. J., de Pillis, L., & De Pillis, L. G. (2015). Linear algebra with applications (pp. 337-350). Boston: Pearson.